凸优化（上）

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主要参考资料：



Convex Optimization
Stephen Boyd and Lieven Vandenberghe

Cambridge University Press

Boyd老爷子直接提供了下载链接。
http://stanford.edu/~boyd/cvxbook/

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Affine and Convex Sets

line and line segments

Suppose $x_1 \neq x_2$ are two points in $\mathbf{R}^n$,那么这样可以定义直线

$$y = \theta x_1 + (1-\theta)x_2$$

当θ取值0~1时，定义的是线段。

Affine sets

A set $C \subseteq \mathbf{R}^n$ is *affine* if the line through any two distinct points in C lies in C

即，affine sets包含了其内点的线性组合，只要系数和为1.

可推广到多个点：

$\theta_1 x_1 + \cdots + \theta_k x_k$, where $\theta_1 + \cdots + \theta_k = 1$（ affine combination ）

offset

若C是affine set，x_0属于C，那么可以如下定义一个set

$$V = C - x_0 = \{x - x_0 \mid x \in C\}$$

V是子空间，对加法和数乘封闭。

Affine sets和线性方程

线性方程组的解集(solution set)是affine set。

每个affine set都是某个线性方程组的解集。

Affine hull

对于任意集合$C \subseteq \mathbf{R^n}$, the set of all affine combinations of points in C 称为the affine hull of C:

$$\mathbf{aff}\, C = \{\theta_1 x_1 + \cdots + \theta_k x_k \mid x_1, \ldots, x_k \in C,\ \theta_1 + \cdots + \theta_k = 1\}$$

A affine hull of C is the smallest affine set that contains C

if S is any affine set with C ⊆ S, then **aff** C ⊆ S.

Affine dimension

We define the affine dimension of a set C as the dimension of its affine hull

例：二维平面上的单位圆的affine dimension是2

Convex set

A set C is *convex* if the line segment between any two points in C lies in C, *i.e.*, if for any $x_1,\ x_2 \in C$ and any θ with $0 \leq \theta \leq 1$, we have

$$\theta x_1 + (1-\theta)x_2 \in C.$$

可以推广到多个：

$$\theta_1 x_1 + \cdots + \theta_k x_k, \text{ where } \theta_1 + \cdots + \theta_k = 1 \text{ and } \theta_i \geq 0,\ i = 1, \ldots, k$$

（convex combination）

affine set都是convex sets.

affine set的θ取任意实数，convex的θ取0~1.

a set is convex if every point in the set can be seen by every other point

Convex hull

The convex hull of a set C, denoted conv C, is the set of all convex combinations of points in C。

$$\mathbf{conv}\, C = \{\theta_1 x_1 + \cdots + \theta_k x_k \mid x_i \in C,\ \theta_i \geq 0,\ i = 1, \ldots, k,\ \theta_1 + \cdots + \theta_k = 1\}$$

A Convex hull of C is the smallest convex set that contains C

Generalization of Convex Concept

Continuous

$$\text{suppose } p : \mathbf{R}^n \to \mathbf{R} \text{ satisfies } p(x) \geq 0 \text{ for all } x \in C \text{ and } \textstyle\int_C p(x)\, dx = 1, \text{ where } C \subseteq \mathbf{R}^n \text{ is convex. Then}$$

$$\int_C p(x) x\, dx \in C,$$

Probabilistic

In the most general form, suppose C ⊆ Rn is convex and x is a random vector with x ∈ C with probability one. Then E x ∈ C. Indeed, this form includes all the others as special cases.

Cones

A set C is called a **cone**, or nonnegative homogeneous, if for every x ∈ C and θ ≥ 0 we have θx ∈ C.

A set C is a **convex cone** if it is convex and a cone, which means that for any x 1 , x 2 ∈ C and θ 1 , θ 2 ≥ 0, we have

$$\theta_1 x_1 + \theta_2 x_2 \in C.$$

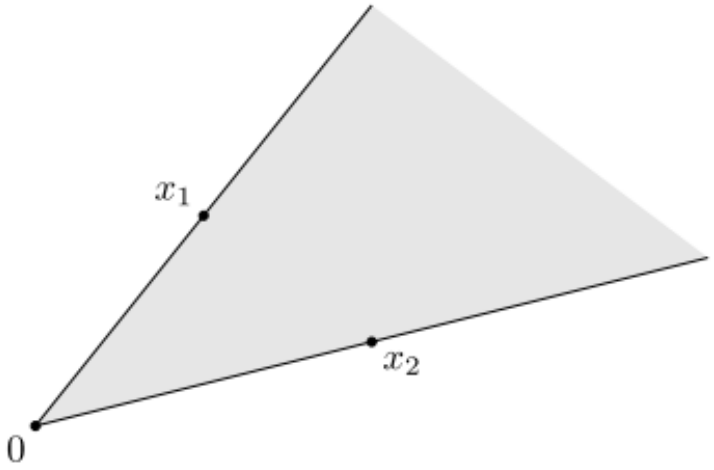


Figure 2.4 The pie slice shows all points of the form $\theta_1 x_1 + \theta_2 x_2$, where $\theta_1,\ \theta_2 \geq 0$. The apex of the slice (which corresponds to $\theta_1 = \theta_2 = 0$) is at 0; its edges (which correspond to $\theta_1 = 0$ or $\theta_2 = 0$) pass through the points x_1 and x_2.

Conic Combination

(work with Convex Cone, not just Cone)
A point of the form $\theta_1 x_1 + \cdots + \theta_k x_k$ with θ 1 , … , θ k ≥ 0 is called a conic combination (or a nonnegative linear combination) of x 1 , … , x k .

If x i are in a convex cone C, then every conic combination of x i is in C.
Conversely, a set C is a Convex cone if and only if it contains all conic combinations of its elements.

Like convex (or affine) combinations, the idea of conic combination can be generalized to infinite sums and integrals.

Conic Hull

The conic hull of a set C is the set of all conic combinations of points in C, i.e.,

$$\{\theta_1 x_1 + \cdots + \theta_k x_k \mid x_i \in C,\ \theta_i \geq 0,\ i = 1, \ldots, k\},$$

It is also the smallest convex cone that contains C

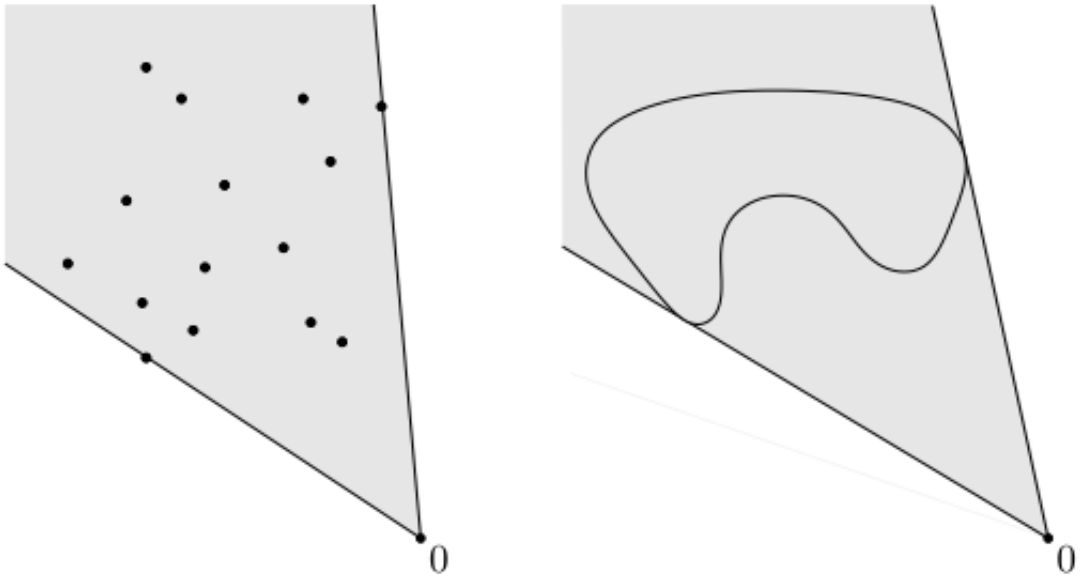


Figure 2.5 The conic hulls (shown shaded) of the two sets of figure 2.3.

Important Examples

- The empty set ∅ , any single point , and the whole space $\boldsymbol{R^n}$ are affine (hence, convex) subsets of $\boldsymbol{R^n}$.
- Any line is affine. If it passes through zero, it is a subspace, hence also a convex cone.
- A line segment is convex, but not affine (unless it reduces to a point).
- A ray, which has the form { x 0 + θv | θ ≥ 0 } , where v 6 = 0, is convex, but not affine. It is a convex cone if its base x 0 is 0.
  • Any subspace is affine, and a convex cone (hence convex).

Hyperplanes and halfspaces

A **hyperplane** is a set of the form

$$\{x \mid a^T x = b\}$$

where a ∈ $\boldsymbol{R^n}$,$\boldsymbol{a \neq 0}$, and b ∈ R.
It is affine, since analytically it is the solution set of a nontrivial linear equation among the components of x

A **hyperplane** divides $\boldsymbol{R^n}$ into two halfspaces. A (closed) halfspace is a set of the form

$$\{x \mid a^T x \leq b\}$$

where $\boldsymbol{a \neq 0}$, i.e., the solution set of one (nontrivial) linear inequality.

Halfspaces are convex, but not affine.

Euclidean balls and ellipsoids

- **ball**
  A *(Euclidean) ball* (or just ball) in $\mathbf{R}^n$ has the form

  $$B(x_c, r) = \{x \mid \|x - x_c\|_2 \leq r\} = \{x \mid (x - x_c)^T(x - x_c) \leq r^2\},$$

  where $r > 0$, and $\|\cdot\|_2$ denotes the Euclidean norm, *i.e.*, $\|u\|_2 = (u^T u)^{1/2}$.

A Euclidean ball is a convex set

- **ellipsoids** are a kind of convex set which have the form
  $\mathcal{E} = \{x \mid (x - x_c)^T P^{-1}(x - x_c) \leq 1\},$
  where P is symmetric and positive definite
- The matrix P determines how far the ellipsoid extends in every direction from x_c
- The lengths of the semi-axes of E are given by $\sqrt{\lambda}$, where λ i are the eigenvalues of P.
- A ball is an ellipsoid with $P = r^2 I$.

Norm balls and norm cones

From the general properties of norms it can be shown that
a **norm ball** of radius r and center x_c , given by $\{x \| \|x - x_c\| \leq r\}$,
is convex.

The **norm cone** associated with the norm $\|.\|$ is the set
$C = \{(x, t) \mid \|x\| \leq t\} \subseteq \mathbf{R}^{n+1}$
it is a convex cone.

Example:

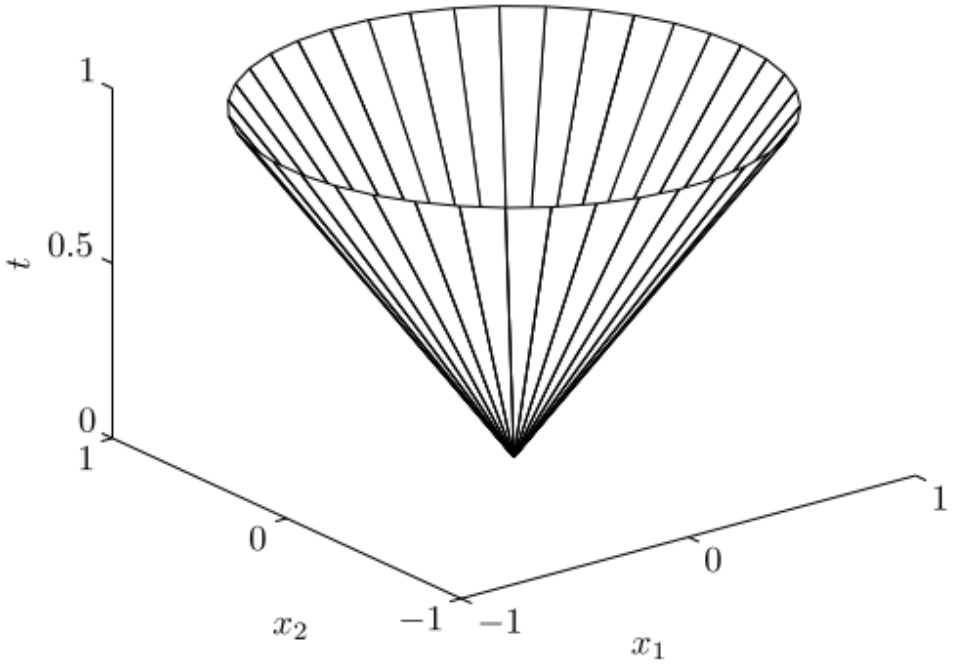


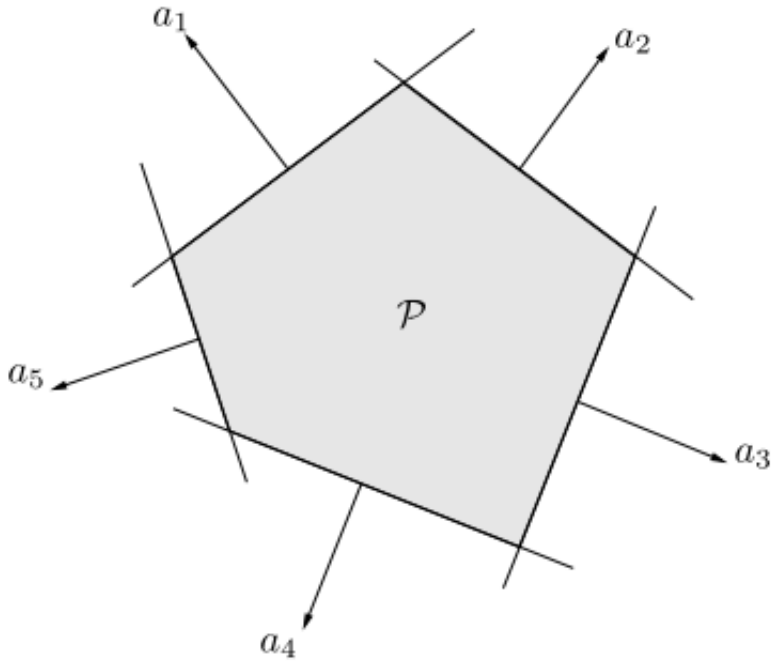
Figure 2.10 Boundary of second-order cone in $\mathbf{R}^3$, $\{(x_1, x_2, t) \mid (x_1^2+x_2^2)^{1/2} \leq t\}$.

polyhedron

A **polyhedron** is defined as the solution set of a finite number of linear equalities and inequalities:

$$\mathcal{P} = \{x \mid a_j^T x \leq b_j,\ j = 1, \ldots, m,\ c_j^T x = d_j,\ j = 1, \ldots, p\}$$

A polyhedron is thus the intersection of a finite number of halfspaces and hyperplanes.



Affine sets (e.g., subspaces, hyperplanes, lines), rays, line segments, and halfspaces are all **polyhedra**.

Simplexes

Suppose the k + 1 points $v_0, \ldots, v_k \in R^n$ are affinely independent, which means $v_1 - v_0, \ldots, v_k - v_0$ are linearly independent. The simplex determined by them is given by

$$C = \mathbf{conv}\{v_0, \ldots, v_k\} = \{\theta_0 v_0 + \cdots + \theta_k v_k \mid \theta \succeq 0,\ \mathbf{1}^T\theta = 1\},$$

Simplexes are a kind of polyhedra.
就是几个点连起来的凸包。
2d：线段
3d：三角形
4d：四面体

Operations that preserve convexity

Intersection $S_1 \cap S_2$

Direct or Cartesian product

$$S_1 \times S_2 = \{(x_1, x_2) \mid x_1 \in S_1,\ x_2 \in S_2\}$$

Affine Function $f(x) = Ax + b$

If S is a convex set,

- $f(S) = \{f(x) \mid x \in S\},$ is convex set
- $f^{-1}(S) = \{x \mid f(x) \in S\},$ is convex set

Examples:

- scaling and translation
  $\alpha S = \{\alpha x \mid x \in S\}, \qquad S + a = \{x + a \mid x \in S\}$
- Projection: $T = \{x_1 \in \mathbf{R}^m \mid (x_1, x_2) \in S \text{ for some } x_2 \in \mathbf{R}^n\}$
- Sum: $S_1 + S_2 = \{x + y \mid x \in S_1,\ y \in S_2\}.$

Generalized inequalities

Proper cones and generalized inequalities

A cone $K \subseteq \mathbf{R}^n$ is called a **proper cone** if it satisfies the following:

- K is convex.
- K is closed.
- K is *solid*, which means it has nonempty interior.
- K is *pointed*, which means that it contains no line (or equivalently, $x \in K,\ -x \in K \implies x = 0$).

A proper cone K can be used to define a **generalized inequality**:

$$x \preceq_K y \iff y - x \in K$$

When $K = \mathbf{R}_+$, the partial ordering $\preceq_K$ is the usual ordering $\leq$ on $\mathbf{R}$, and the strict partial ordering $\prec_K$ is the same as the usual strict ordering $<$ on $\mathbf{R}$.

Separating and supporting hyperplanes

Separating hyperplane theorem

Suppose C and D are nonempty disjoint convex sets, i.e., C ∩ D = ∅ . Then there exist $a \neq 0$ and b such that $a^T x \leq b$ for all x ∈ C and $a^T x \geq b$ for all x ∈ D. In other words, the affine function $a^T x - b$ is nonpositive on C and nonnegative on D. The hyperplane { x | $a^T x = b$ } is called a **separating hyperplane** for the sets C and D, or is said to separate the sets C and D.

Supporting hyperplane

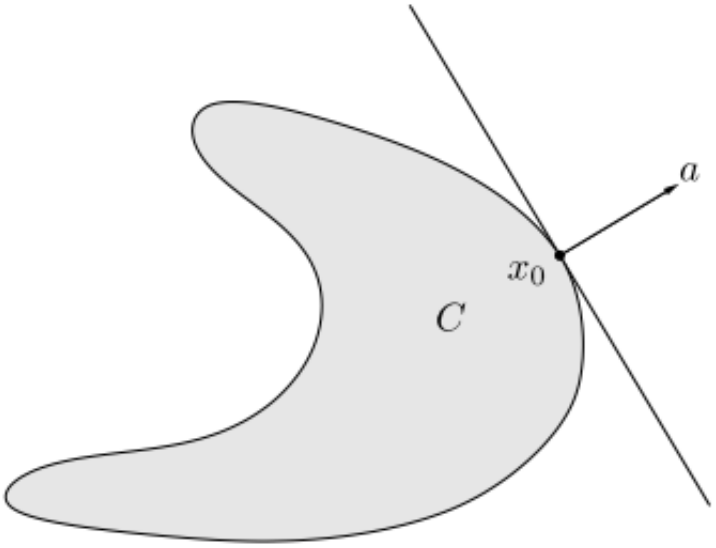


Figure 2.21 The hyperplane $\{x \mid a^T x = a^T x_0\}$ supports C at x_0.

Dual cones and generalized inequalities

K is a cone. The **dual cone** of K is

$$K^* = \{y \mid x^T y \geq 0 \text{ for all } x \in K\}$$

K^* is a cone, and it is convex even when K is not convex.

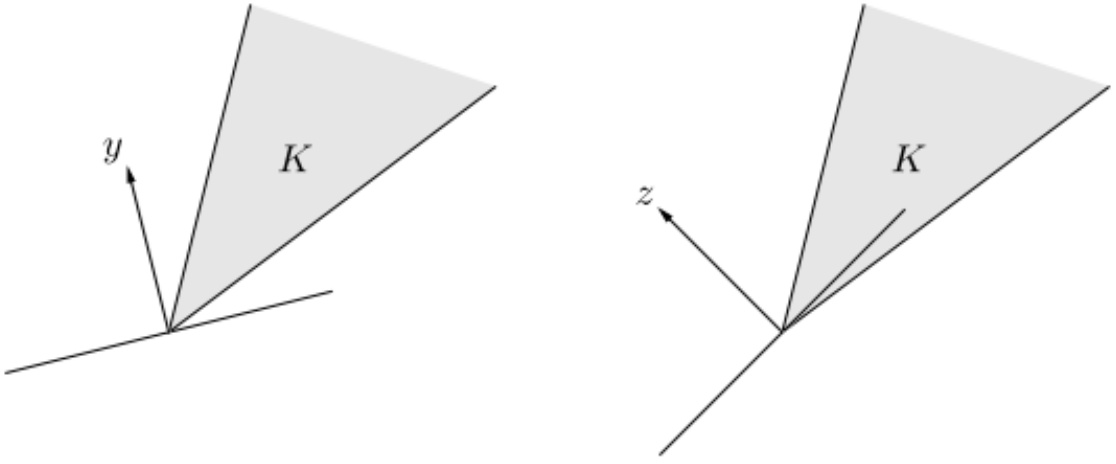


Figure 2.22 *Left.* The halfspace with inward normal y contains the cone K, so $y \in K^*$. *Right.* The halfspace with inward normal z does not contain K, so $z \notin K^*$.

Some feature of dual-cone

- K^* is closed and convex.
- $K_1 \subseteq K_2$ implies $K_2^* \subseteq K_1^*$.
- If K has nonempty interior, then K^* is pointed.
- If the closure of K is pointed then K^* has nonempty interior.
- K^{**} is the closure of the convex hull of K. (Hence if K is convex and closed, $K^{**} = K$.)

02-Convex Functions

02-Convex Functions
- Basic Properties
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    - Scalar composition
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    - Definition
    - Differentiable f
    - Examples
  - Basic Properties
    - Fenchel’s inequality
    - Conjugate of the conjugate
    - affine transformation
    - Sums of independent functions

Basic Properties

Definition

A function $f : \mathbf{R}^n \to \mathbf{R}$ is *convex* if $\mathbf{dom}\, f$ is a convex set and if for all x, $y \in \mathbf{dom}\, f$, and θ with $0 \leq \theta \leq 1$, we have

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y). \tag{3.1}$$

Geometrically, this inequality means that the line segment between $(x, f(x))$ and $(y, f(y))$, which is the *chord* from x to y, lies above the graph of f (figure 3.1). A function f is *strictly convex* if strict inequality holds in (3.1) whenever $x \neq y$ and $0 < \theta < 1$. We say f is *concave* if $-f$ is convex, and *strictly concave* if $-f$ is strictly convex.

For an **affine function** we always have equality in (3.1), so all affine (and therefore also linear) functions are **both convex and concave**. Conversely, any function that is convex and concave is affine.

A judgement
A function is convex if and only if it is convex when restricted to any line that intersects its domain. In other words f is convex if and only if for all x $\in$ dom f and all v, the function g(t) = f(x + tv) is convex (on its domain, { t | x + tv $\in$ dom f }).

This property is very useful, since it allows us to check whether a function is convex by restricting it to a line.

First Order Condition

Suppose f is differentiable. Then f is convex if and only if dom f is convex and

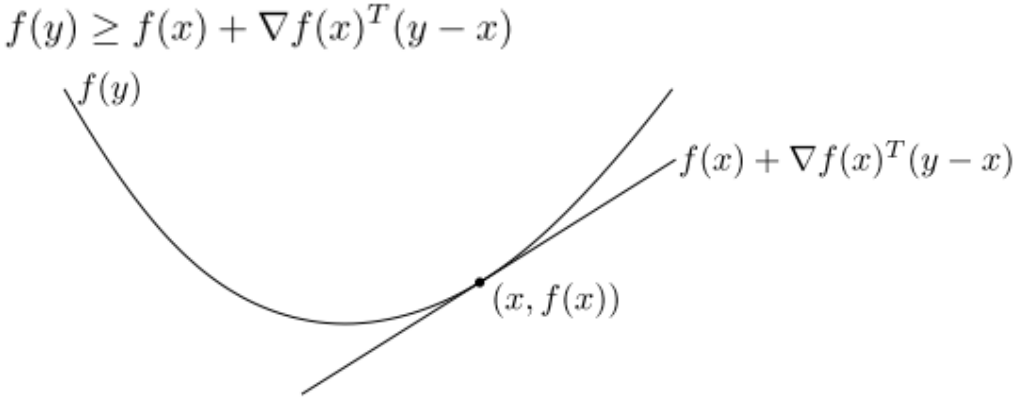


Figure 3.2 If f is convex and differentiable, then $f(x)+\nabla f(x)^T(y-x) \leq f(y)$ for all $x,\ y \in \mathbf{dom}\, f$.

Second Order Condition

We now assume that f is twice differentiable, then f is convex if and only if dom f is convex and its Hessian is positive semidefinite: for all x $\in$ dom f,

$$\nabla^2 f(x) \succeq 0.$$

Some examples

- all linear and affine functions are convex (and concave),
- Norms. Every norm on R^n is convex.
- Max function. f(x) = max { x1 , … , xn } is convex on R^n
- Quadratic.

  Example 3.2 *Quadratic functions.* Consider the quadratic function $f : \mathbf{R}^n \to \mathbf{R}$, with $\mathbf{dom}\, f = \mathbf{R}^n$, given by

  $$f(x) = (1/2)x^T Px + q^T x + r,$$

  with $P \in \mathbf{S}^n$, $q \in \mathbf{R}^n$, and $r \in \mathbf{R}$. Since $\nabla^2 f(x) = P$ for all x, f is convex if and only if $P \succeq 0$ (and concave if and only if $P \preceq 0$).

  For quadratic functions, strict convexity is easily characterized: f is strictly convex if and only if $P \succ 0$ (and strictly concave if and only if $P \prec 0$).

-
  - *Exponential.* e^{ax} is convex on $\mathbf{R}$, for any $a \in \mathbf{R}$.
  - *Powers.* x^a is convex on $\mathbf{R}_{++}$ when $a \geq 1$ or $a \leq 0$, and concave for $0 \leq a \leq 1$.
  - *Powers of absolute value.* $|x|^p$, for $p \geq 1$, is convex on $\mathbf{R}$.
  - *Logarithm.* $\log x$ is concave on $\mathbf{R}_{++}$.

Sublevel sets

$$C_\alpha = \{x \in \mathbf{dom}\, f \mid f(x) \leq \alpha\}$$

Sublevel sets of a convex function are convex, for any value of α.

Epigraph

The **graph** of a function $f : \mathbf{R}^n \to \mathbf{R}$ is

$$\{(x, f(x)) \mid x \in \mathbf{dom}\, f\},$$

The **epigraph** of a function is

$$\mathbf{epi}\, f = \{(x,t) \mid x \in \mathbf{dom}\, f,\ f(x) \le t\}$$

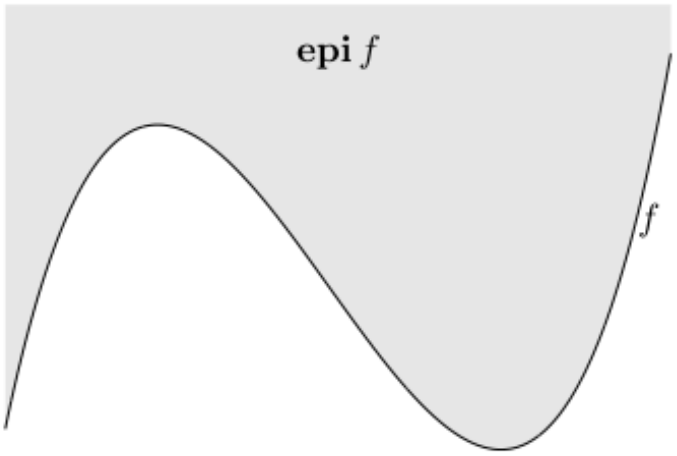


Figure 3.5 Epigraph of a function f, shown shaded. The lower boundary, shown darker, is the graph of f.

The link between convex sets and convex functions is via the epigraph: A function is convex if and only if its epigraph is a convex set.

Operations that preserve convexity

Nonnegative weighted sums

$$f = w_1 f_1 + \cdots + w_m f_m$$

Composition with an affine mapping

$$g(x) = f(Ax + b),$$

f and g have the same convexity

Pointwise maximum and supremum

If f1 and f2 are convex functions then their pointwise maximum f, defined by

$$f(x) = \max\{f_1(x), f_2(x)\},$$

with dom f = dom f 1 ∩ dom f 2 , is also convex

This could easily extend:

$$f(x) = \max\{f_1(x), \ldots, f_m(x)\}$$

Composition

$$f(x) = h(g(x)), \qquad \mathbf{dom}\, f = \{x \in \mathbf{dom}\, g \mid g(x) \in \mathbf{dom}\, h\}.$$

Scalar composition

f is convex if h is convex, $\tilde{h}$ is nondecreasing, and g is convex,
f is convex if h is convex, $\tilde{h}$ is nonincreasing, and g is concave,
f is concave if h is concave, $\tilde{h}$ is nondecreasing, and g is concave,
f is concave if h is concave, $\tilde{h}$ is nonincreasing, and g is convex.

Here $\tilde{h}$ denotes the extended-value extension of the function h, which assigns the value ∞ (−∞) to points not in dom h for h convex (concave).

Vector composition

Use the second order condition:

$$f''(x) = g'(x)^T \nabla^2 h(g(x)) g'(x) + \nabla h(g(x))^T g''(x),$$

we could derive many rules:

f is convex if h is convex, h is nondecreasing in each argument, and g_i are convex,
f is convex if h is convex, h is nonincreasing in each argument, and g_i are concave,
f is concave if h is concave, h is nondecreasing in each argument, and g_i are concave.

Perspective of a function

If $f : \mathbf{R}^n \to \mathbf{R}$, then the *perspective* of f is the function $g : \mathbf{R}^{n+1} \to \mathbf{R}$ defined by

The Conjugate Function

Definition&examples

Definition

Let $f : \mathbf{R}^n \to \mathbf{R}$. The function $f^* : \mathbf{R}^n \to \mathbf{R}$, defined as

$$f^*(y) = \sup_{x \in \mathbf{dom}\, f} \left(y^T x - f(x)\right), \tag{3.18}$$

is called the *conjugate* of the function f. The domain of the conjugate function consists of $y \in \mathbf{R}^n$ for which the supremum is finite, *i.e.*, for which the difference $y^T x - f(x)$ is bounded above on $\mathbf{dom}\, f$. This definition is illustrated in figure 3.8

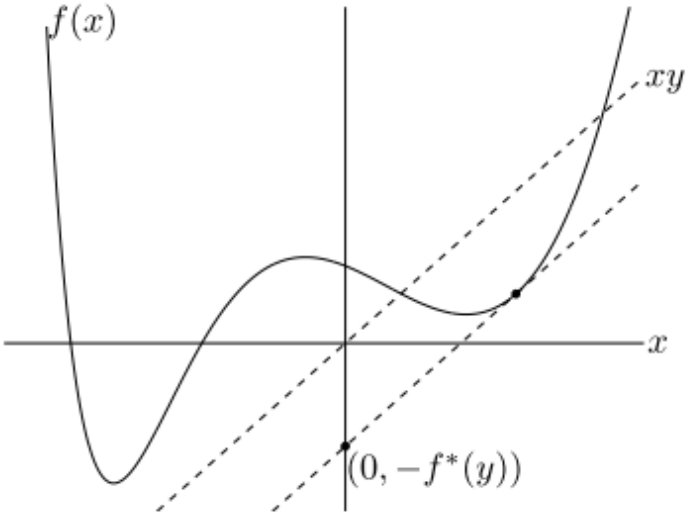


Figure 3.8 A function $f : \mathbf{R} \to \mathbf{R}$, and a value $y \in \mathbf{R}$. The conjugate function $f^*(y)$ is the maximum gap between the linear function yx and $f(x)$, as shown by the dashed line in the figure. If f is differentiable, this occurs at a point x where $f'(x) = y$.

f^* is convex, whether or not f is convex.

Differentiable f

The conjugate of a differentiable function f is also called the *Legendre transform* of f.

Let $z \in R^n$ be arbitrary and define $y = \nabla f(z)$.

Then we have

$$f^*(y) = z^T \nabla f(z) - f(z).$$

Examples

Affine function. $f(x) = ax + b$. As a function of x, $yx - ax - b$ is bounded if and only if $y = a$, in which case it is constant. Therefore the domain of the conjugate function f^* is the singleton $\{a\}$, and $f^*(a) = -b$.

Example 3.22 *Strictly convex quadratic function.* Consider $f(x) = \frac{1}{2}x^TQx$, with $Q \in \mathbf{S}^n_{++}$. The function $y^Tx - \frac{1}{2}x^TQx$ is bounded above as a function of x for all y. It attains its maximum at $x = Q^{-1}y$, so

$$f^*(y) = \frac{1}{2}y^TQ^{-1}y.$$

Basic Properties

Fenchel's inequality

From the definition of conjugate function, we immediately obtain the inequality

$$f(x) + f^*(y) \geq x^Ty$$

for all x, y. This is called *Fenchel's inequality* (or *Young's inequality* when f is differentiable).

For example with $f(x) = (1/2)x^TQx$, where $Q \in \mathbf{S}^n_{++}$, we obtain the inequality

$$x^Ty \leq (1/2)x^TQx + (1/2)y^TQ^{-1}y.$$

Conjugate of the conjugate

if f is convex, and f is closed (i.e., **epi f** is a closed set), then $f^{**} = f$

affine transformation

- $af(x) + b$: For $a > 0$ and $b \in R$, the conjugate of $g(x) = af(x) + b$ is $g^*(y) = af^*(y/a) - b$.
- $f(Ax + b)$:Suppose $A \in R^{n\times n}$ is nonsingular and $b \in R^n$. Then the conjugate of $g(x) = f(Ax + b)$ is,with dom g^* = A^T dom f^*.

Sums of independent functions

03-Convex Optimization Problems

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & a_i^T x = b_i, \quad i = 1, \ldots, p, \end{array}$$

03-Convex Optimization Problems
- Definition
- An optimality criterion for differentiable
  - Unconstrained problems
  - Problems with equality constraints only

Definition

$f_0, \ldots, f_m$ are convex functions.

Or in other words,

- the objective function must be convex,
- the inequality constraint functions must be convex,
- the equality constraint functions $h_i(x) = a^T - b_i$ must be affine.

The feasible set of a convex optimization problem is convex, since it is the intersection of some convex sets.

In a convex optimization problem, we minimize a convex objective function over a convex set.

An optimality criterion for differentiable

Generally speaking, for any convex optimization problem that is differentiable,
x is optimal if and only if $x \in X$ and

$$\nabla f_0(x)^T (y - x) \geq 0 \text{ for all } y \in X.$$

Where X denotes the feasible set of the problem.
Next we talk about more specific situations.

Unconstrained problems

The optimality condition reduce to the necessary and sufficient condition:

$$\nabla f_0(x) = 0$$

Example:Unconstrained quadratic optimization

$$f_0(x) = (1/2)x^T Px + q^T x + r,$$

where $P \in \mathbf{S}_+^n$ (which makes f_0 convex).

The necessary and sufficient condition for x to be a minimizer:

$$\nabla f_0(x) = Px + q = 0.$$

Possible cases:

- If $q \notin \mathcal{R}(P)$, then there is no solution. In this case f_0 is unbounded below.
- If $P \succ 0$ (which is the condition for f_0 to be strictly convex), then there is a unique minimizer, $x^\star = -P^{-1}q$.
- If P is singular, but $q \in \mathcal{R}(P)$, then the set of optimal points is the (affine) set $X_{\text{opt}} = -P^\dagger q + \mathcal{N}(P)$, where $P^\dagger$ denotes the pseudo-inverse of P (see §A.5.4).

Problems with equality constraints only

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & Ax = b. \end{array}$$

Deduction:

Here the feasible set is affine. We assume that it is nonempty; otherwise the problem is infeasible. The optimality condition for a feasible x is that

$$\nabla f_0(x)^T (y - x) \geq 0$$

must hold for all y satisfying $Ay = b$. Since x is feasible, every feasible y has the form $y = x + v$ for some $v \in \mathcal{N}(A)$. The optimality condition can therefore be expressed as:

$$\nabla f_0(x)^T v \geq 0 \text{ for all } v \in \mathcal{N}(A).$$

If a linear function is nonnegative on a subspace, then it must be zero on the subspace, so it follows that $\nabla f_0(x)^T v = 0$ for all $v \in \mathcal{N}(A)$. In other words,

$$\nabla f_0(x) \perp \mathcal{N}(A).$$

Using the fact that $\mathcal{N}(A)^\perp = \mathcal{R}(A^T)$, this optimality condition can be expressed as $\nabla f_0(x) \in \mathcal{R}(A^T)$, *i.e.*, there exists a $\nu \in \mathbf{R}^p$ such that

$$\nabla f_0(x) + A^T \nu = 0.$$

04-Duality

04-Duality
- Definition
  - Lagrangian
  - Lagrange Dual Function
    - Lower bounds
    - Relation with conjugate function
  - The Lagrange dual problem
- Weak Duality
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Definition

Consider a standard form optimization problem:

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p, \end{array}$$

$x \in R^n$.
We assume the feasible domain is not empty,
denote the optimal value by p^*,
and do **not** assume the problem is convex.

Lagrangian

We define the *Lagrangian* L: $\mathbf{R}^n \times \mathbf{R}^m \times \mathbf{R}^p \to \mathbf{R}$ of the problem as:

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x),$$

with $\mathbf{dom}\, L = \mathcal{D} \times \mathbf{R}^m \times \mathbf{R}^p$.

$\mathcal{D} = \bigcap_{i=0}^{m} \mathbf{dom}\, f_i \ \cap \ \bigcap_{i=1}^{p} \mathbf{dom}\, h_i$

- we refer to λ_i as **Lagrange multiplier** associated with the ith **inequality constraint.**
- we refer to ν_i as **Lagrange multiplier** associated with the ith **equality constraint**.
- The vectors λ and ν are called the **dual variables** or **Lagrange multiplier vectors** associated with the problem

Convexity: Analysis for yourself. For example, when the problem is convex, and $\lambda >= 0$, then L as a function of x is the sum of three convex terms, thus it is convex.

Lagrange Dual Function

We define the *Lagrange dual function* (or just *dual function*) $g : \mathbf{R}^m \times \mathbf{R}^p \to \mathbf{R}$ as the minimum value of the Lagrangian over x: for $\lambda \in \mathbf{R}^m$, $\nu \in \mathbf{R}^p$,

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) = \inf_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x) \right).$$

When the Lagrangian is unbounded below in x, the dual function takes on the value $-\infty$. Since the dual function is the pointwise infimum of a family of affine functions of (λ, ν), it is concave, even when the problem (5.1) is not convex.

Lower bounds

The dual function yields lower bounds on the optimal value $p^\star$ of the problem:

For any $\lambda \succeq 0$ and any ν we have

$$g(\lambda, \nu) \leq p^\star.$$

This is easily proved as follows: for any point $\bar{x}$ that is feasible for the prime problem,

$$g(\lambda, \upsilon)) = \inf_{x \in D} L(x, \lambda, \upsilon) \leq L(\bar{x}, \lambda, \upsilon) \leq f(\bar{x})$$

This lower bounds only holds when $\text{only when } \lambda \succeq 0 \text{ and } (\lambda, \nu) \in \mathbf{dom}\, g,$.

$\lambda > 0$ is necessary for the deduction;

$(\lambda, \nu) \in$ dom g ensure the value is meaningful(not $-\infty$)

Relation with conjugate function

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & Ax \preceq b \\ & Cx = d. \end{array} \tag{5.10}$$

Using the conjugate of f_0 we can write the dual function for the problem (5.10) as

$$\begin{aligned} g(\lambda,\nu) &= \inf_x \left(f_0(x) + \lambda^T(Ax-b) + \nu^T(Cx-d)\right) \\ &= -b^T\lambda - d^T\nu + \inf_x \left(f_0(x) + (A^T\lambda + C^T\nu)^T x\right) \\ &= -b^T\lambda - d^T\nu - f_0^*(-A^T\lambda - C^T\nu). \end{aligned} \tag{5.11}$$

The domain of g follows from the domain of f_0^*:

$$\mathbf{dom}\, g = \{(\lambda,\nu) \mid -A^T\lambda - C^T\nu \in \mathbf{dom}\, f_0^*\}.$$

The Lagrange dual problem

The Lagrange dual function gives us a lower bound on the optimal value $p^\star$ of the optimization problem depends on some parameters λ, ν.
A natural question is: What is the best lower bound that can be obtained from the Lagrange dual function?
This is a new optimization problem called ***Lagrange dual problem*** associated with the original problem.

$$\begin{array}{ll} \text{maximize} & g(\lambda,\nu) \\ \text{subject to} & \lambda \succeq 0. \end{array}$$

- The original problem is called prime problem,
- a pair (λ,ν) with $\lambda \succeq 0$ and $g(\lambda,\nu) > -\infty$, is called ***dual feasible***, which means it is feasible for the dual problem.
- We refer to $(\lambda^\star, \nu^\star)$ as dual optimal or optimal Lagrange multipliers if they are optimal for the dual problem.
- The Lagrange dual problem is a **convex** optimization problem whether or not the primal problem is convex.

Weak Duality

The optimal value of the Lagrange dual problem, which we denote $d^\star$, is, by definition, the best lower bound on $p^\star$ that can be obtained from the Lagrange dual function. In particular, we have the simple but important inequality

$$d^\star \le p^\star, \tag{5.23}$$

which holds even if the original problem is not convex. This property is called *weak duality*.

- The weak duality inequality holds when $d^\star$ and $p^\star$ are infinite. For example, if the primal problem is unbounded below, so that $p^\star = -\infty$, we must have $d^\star = -\infty$
- We refer to the difference $p^\star - d^\star$ as the ***optimal duality gap*** of the original problem
- The gap sometimes is used to find a lower bound on the optimal value of a problem that is difficult to solve, since the dual problem is always convex
- More generally, a feasible point x of prime problem and a feasible point of dual problem (λ, ν) could give a **duality gap** $f_0(x) - g(\lambda, \nu)$, which indicates the range of possible solution, and often used as a stopping criteria for optimization algorithms.

Strong Duality

If the optimal duality gap is zero, ie

$$d^\star = p^\star$$

The condition is called ***strong duality***

It is not always hold.
One condition that could result into strong duality is ***Slater's condition***

Slater's condition

Here we only discuss prime problems whose equality constraints are linear.
Define the relative interior of set C as

$$\mathbf{relint}\, C = \{x \in C \mid B(x, r) \cap \mathbf{aff}\, C \subseteq C \text{ for some } r > 0\}$$

Slater's condition says:

> There exists an $x \in \mathbf{relint}\, D$ such that
> $$f_i(x) < 0, \quad i = 1, \ldots, m, \qquad Ax = b.$$

Such a point is sometimes called ***strictly feasible***, since the inequality constraints hold with strict inequalities.

When some of the inequality constraint functions are affine, the condition could be weaker:
If the first k constraint functions f1 , … , fk are affine,

> There exists an $x \in \mathbf{relint}\, D$ such that
> $$f_i(x) \leq 0, \quad i = 1, \ldots, k, \qquad f_i(x) < 0, \quad i = k+1, \ldots, m, \qquad Ax = b.$$

In other words, the affine inequalities do not need to hold with strict inequality

Here comes the ***slater's theorem***:

> **The strong duality holds, if Slater's condition holds and the problem is convex**

Note that nonconvex prime problem could possibly have strong duality. For example,

> strong duality holds for any optimization problem with quadratic objective and one quadratic inequality constraint, provided Slater's condition holds

Optimality Conditions

We do not assume the prime problem is convex, unless explicitly stated.

Note the difference between slater's condition and optimality condition:

- Slater's condition judge a problem;
- Optimality condition judge a feasible point.

Suppose that the primal and dual optimal values are attained and equal (so, strong duality holds), then we could write:

$$\begin{aligned} f_0(x^\star) &= g(\lambda^\star, \nu^\star) \\ &= \inf_x \left(f_0(x) + \sum_{i=1}^m \lambda_i^\star f_i(x) + \sum_{i=1}^p \nu_i^\star h_i(x) \right) \\ &\leq f_0(x^\star) + \sum_{i=1}^m \lambda_i^\star f_i(x^\star) + \sum_{i=1}^p \nu_i^\star h_i(x^\star) \\ &\leq f_0(x^\star). \end{aligned}$$

which means the two inequalities are equalities. This draws two conclusion:

- x^* is a minimizer of $L(x, \lambda^*, \nu^*)$
- $\lambda_i^\star f_i(x^\star) = 0, \quad i = 1, \ldots, m$

Which leads to the following two important conclusions respectively.

Complementary slackness

The second condition above is the complementary slackness.
It could be expressed as

$$\lambda_i^\star > 0 \implies f_i(x^\star) = 0,$$

or

$$f_i(x^\star) < 0 \implies \lambda_i^\star = 0.$$

KKT optimality condition

Assume that the functions f0 , … , fm , h1 , … , hp are differentiable.
The first condition above (x^* is a minimizer of $L(x, \lambda^*, \nu^*)$) means that its gradient must vanish at x^*

$$\nabla f_0(x^\star) + \sum_{i=1}^{m} \lambda_i^\star \nabla f_i(x^\star) + \sum_{i=1}^{p} \nu_i^\star \nabla h_i(x^\star) = 0.$$

This along with other conditions are called the Karush-Kuhn-Tucker (KKT) conditions:

$$\begin{array}{rcll} f_i(x^\star) & \leq & 0, & i = 1, \ldots, m \\ h_i(x^\star) & = & 0, & i = 1, \ldots, p \\ \lambda_i^\star & \geq & 0, & i = 1, \ldots, m \\ \lambda_i^\star f_i(x^\star) & = & 0, & i = 1, \ldots, m \\ \nabla f_0(x^\star) + \sum_{i=1}^m \lambda_i^\star \nabla f_i(x^\star) + \sum_{i=1}^p \nu_i^\star \nabla h_i(x^\star) & = & 0, & \end{array}$$

We got a theorem:

> If strong duality holds for an optimization problem with differentiable objective and constraint functions,
> then any pair of primal and dual optimal points must satisfy the KKT conditions

KKT for convex problem

If the prime problem is convex, then the inverse of KKT condition is also true:

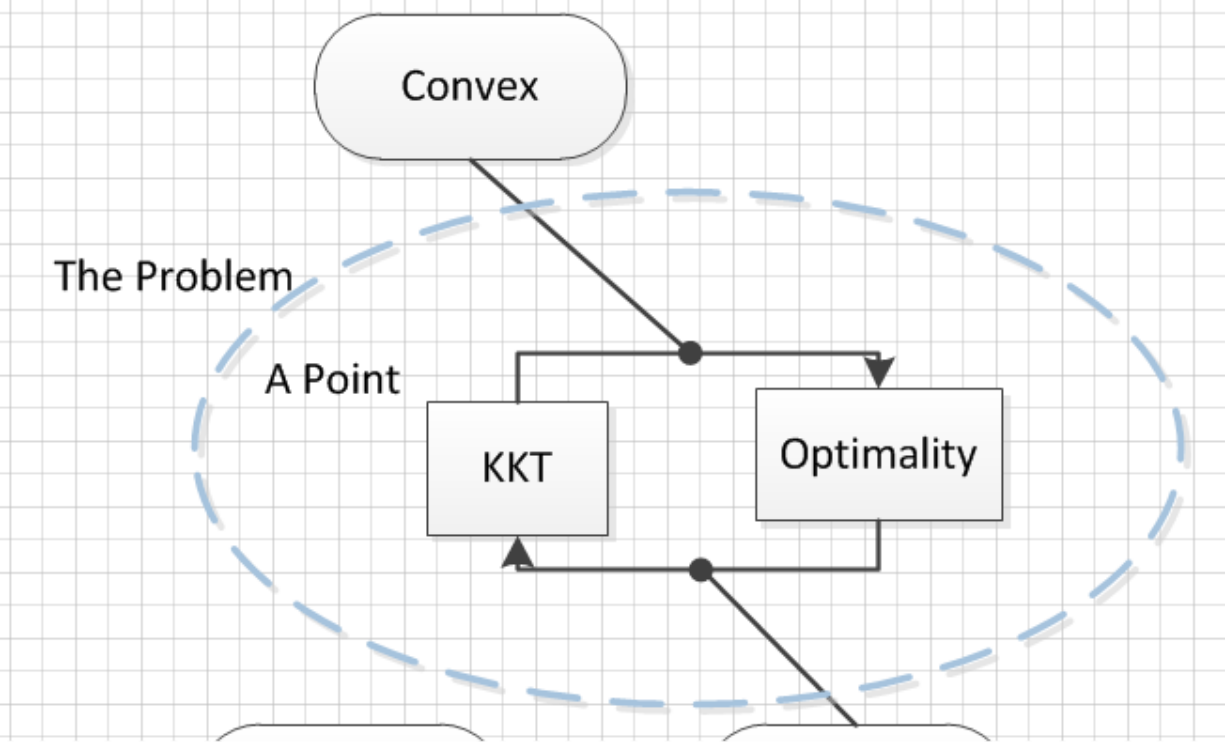
> If a pair of primal and dual pointsx, λ, ν satisfiy the KKT condition for a convex problem,
> then they are prime and dual optimal, with zero duality gap.

KKT for convex problem Plus Slater’s condition

Combining the above two theorem,
we get KKT for a convex problem that satisfies Slater's condition:
(Note that Slater's condition -> strong duality)

> If a convex optimization problem with differentiable objective and constraint functions satisfies Slater's condition,
> then the KKT conditions provide necessary and sufficient conditions for optimality.

The theorems in this chapter could be illustrated as follows:



The KKT conditions play an important role in optimization. In a few special cases it is possible to solve the KKT conditions (and therefore, the optimization problem) analytically.

More generally, many algorithms for convex optimization are conceived as, or can be interpreted as, methods for solving the KKT conditions.

Examples

QP with equation constraints

Example 5.1 *Equality constrained convex quadratic minimization.* We consider the problem

$$\begin{array}{ll} \text{minimize} & (1/2)x^T Px + q^T x + r \\ \text{subject to} & Ax = b, \end{array} \tag{5.50}$$

where $P \in \mathbf{S}_+^n$. The KKT conditions for this problem are

$$Ax^\star = b, \qquad Px^\star + q + A^T\nu^\star = 0,$$

which we can write as

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^\star \\ \nu^\star \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}.$$

Solving this set of m + n equations in the m + n variables $x^\star, \nu^\star$ gives the optimal primal and dual variables for (5.50).